Effects of Pilot Fuel and Liner Cooling on the Flame Structure in a Full Scale Swirl-Stabilized Combustion Setup

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The flame structure and the limits of operation of a lean premixed swirl flame were experimentally investigated under piloted and nonpiloted conditions. Flame stabilization and blow out limits are discussed with respect to pilot fuel injection and combustor liner cooling for lean operating conditions. Two distinctly different flow patterns are found to develop depending on piloting and liner cooling parameters. These flow patterns are characterized with respect to flame stability, blow out limits, combustion noise, and emissions. The combustion system explored consists of a single burner similar to the burners used in Siemens annular combustion systems. The burner feeds a distinctively nonadiabatic combustion chamber operated with natural gas under atmospheric pressure. Liner cooling is mimicked by purely convective cooling and an additional flow of "leakage air" injected into the combustion chamber. Both additional air flow and the pilot fuel ratio were found to have a strong influence on the flow structure and stability of the flame close to the lean blow off (LBO) limit. It is shown by laser Doppler velocimetry that the angle of the swirl cone is strongly affected by pilot fuel injection. Two distinct types of flow patterns are observed close to LBO in this large scale setup: While nonpiloted flames exhibit tight cone angles and small inner recirculation zones (IRZs), sufficient piloting results in a wide cone angle and a large IRZ. Only in the latter case, the main flow becomes attached to the combustor liner. Flame structures deduced from flow fields and CH-chemiluminescence images depend on both the pilot fuel injection and liner cooling. [DOI: 10.1115/1.4000588]

1 Introduction

Lean premixed swirl stabilized combustion is successfully applied in gas turbine combustion systems to maintain low NO_x and CO-emissions over a wide operating range. The flow patterns in these combustions systems are characterized by a conical main flow zone and an internal recirculation zone in which hot burnt gases are recirculated. Stable combustion over a wide operating range is ensured by the turbulent reacting shear layer formed between the hot recirculating gases and the unburnt fuel/air mixture of the main flow. Extensive work was performed to understand swirl flows and their effects in combustion processes. The work of Gupta et al. [1] gives a comprehensive overview, while Meier et al. [2] and Beck et al. [3] include recent advancements. Requirements to modern heavy duty gas turbine systems are low pollutant emissions, safe operation, and a wide operating range. While low NO_x emissions require air/fuel mixtures close to the lean blow off limit, the risks of thermo-acoustic instabilities and lean blow off often can be avoided by choosing richer mixtures.

In the light of this contradiction, multifuel staged systems have been introduced to increase the operating range of lean premixed combustion systems. In the burners used in the annular combustion chambers of Siemens heavy duty gas turbine series (Fig. 1) a lean premixed main fuel stage is combined with a central "pilot" fuel stage, which provides a locally enriched air/fuel mixture at the center of the burner outlet. This system extends the operating range of the swirl flame toward leaner conditions [5,6]. The properties of combustion systems based on this staged burner concept were investigated at various levels of abstraction from field tests [7] to atmospheric single burner testing [8] and from simulations covering specific details, such as mixing [9], to extensive large eddy simulations including the entire combustion system [10].

Although many aspects of swirl stabilized lean combustion systems are known today, the effects of pilot fuel injection on the overall flow field of this exemplary lean premixed combustion system are not fully understood. Even though emission measurements taken during field tests [7] clearly indicate a potential to reduce NO_x emissions considerably by a reduction in the pilot fuel ratio, the mechanism, which produces the stabilizing effect of the pilot, is yet to be clearly identified. Previous experimental studies using a quasi-adiabatic atmospheric combustion chamber were focused on the mechanisms leading to thermo-acoustic instabilities under conditions richer than the design point [8,11–13]. These investigations indicated that the pilot neither had a measurable effect on the thermo acoustic behavior nor on the lean blow off limits.

In this context, a large scale test rig for a single staged burner was designed, which has features similar to the quasi-adiabatic rig studied earlier. The new setup includes a nonadiabatic liner configuration in order to bring the test case closer to the real engine. The nonadiabatic test rig exhibited the desired effects of pilot fuel injection [14]. In this paper, detailed measurements of the flow field inside the nonadiabatic combustion chamber will be presented in order to highlight some of the effects of pilot fuel injection. Wall heat flux, noise, and emission measurements will be used to link these effects to system performance.

2 Experimental Apparatus

The current setup consists of one staged burner mounted in an atmospheric combustion chamber. Figure 2 shows a sketch of the burner, the combustion chamber, and the corresponding coordi-

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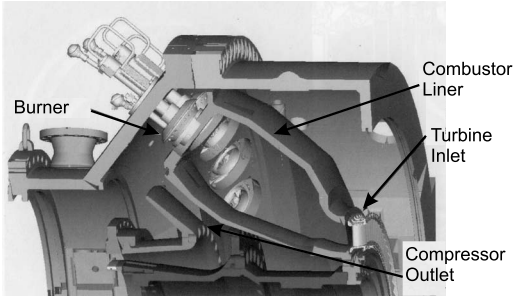

Fig. 1 Annular combustion chamber of the Siemens SGTxF type [4]

nate system. The burner consists of two axis-symmetric swirlers. The majority of the combustion air mass flow $\dot{m}_{B,a}$ passes the diagonal swirler where the main *premix* fuel flow is added to the combustion air. The reactants are intensively mixed before entering the combustion chamber [15]. This main fuel stage provides a lean premixed swirl stabilized main flame with low NO_x emissions [7]. In the order of 13% of the combustion air passes the axial swirler to form a corotating inner vortex. Pilot fuel ($\dot{m}_{P,f}$) can be injected into this inner flow. For operating conditions close to the design point of the gas turbine, the main premixed stage is stabilized using typically 2…10% of the overall fuel flow as pilot fuel. Details about load modulation of the gas turbine in the field are available in the literature [5,7]. The combustion chamber has inner dimensions, as studied previously [11]: A cuboid combustion chamber with square cross section is terminated by a truncated pyramid, which forms the exhaust gas exit. The edge length of the cross section is ≈ 3 times the burner outer diameter d_b. Its surface area is similar to the surface area per burner available in the equivalent gas turbine [8]. In the same sense, the length of the combustion chamber is $\approx 6 \cdot d_b$ in order to match the volume of the test section to 1/24 of the volume of the annular combustion chamber.

Previous work [8,12] used an entirely ceramic liner resulting in quasi-adiabatic conditions in the combustion chamber. Wall surface temperatures were in the order of 90% of the adiabatic flame temperature [9]. This quasi-adiabatic test rig did not show any dependencies of the stability margins on the pilot fuel mass flow under conditions close to lean blow off (LBO) [16]. As the impact of pilot fuel, as seen in gas turbines in the field, was to be studied in the present campaign, the boundary conditions associated to the combustor liner were altered to represent the gas turbine conditions more closely. The liner of the combustion chamber now consists of three functional parts: ceramic surfaces, a metallic liner structure providing a distinct heat flux out of the combustion chamber and a so-called "leakage air" system which allows to inject cold air into the outer recirculation zone (ORZ).

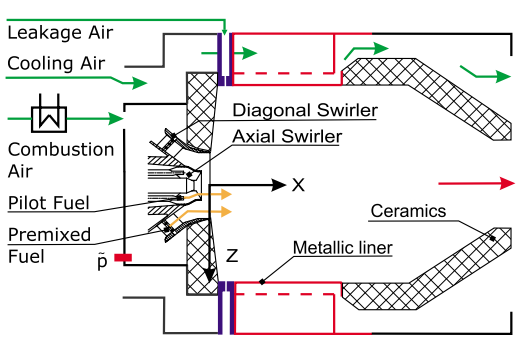

Fig. 2 Schematic of burner and test rig

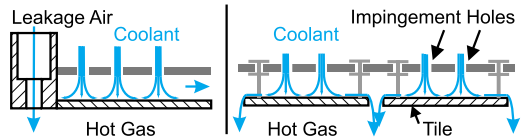

Fig. 3 Schematic of the cooled liner structures of this test setup (left) and a common gas turbine (right)

The metallic liner was introduced to make the present combustion chamber distinctively nonadiabatic. The liner structure is cooled using impinging jets on the outer surface. It stretches around the whole perimeter and is $\approx 2.3 \cdot d_b$ long. The liner temperature was found to be in the order of $\approx 40-70\%$ of the nominal adiabatic flame temperature T_{ref}. The heat flux past the metallic liner was found to vary between 7% and 15% of the nominal thermal power Q_{ref} depending on the combustor load. The coolant flow of the liner was sealed against the combustor. This setup was preferred to convectively cooled tiles found in common gas turbine designs [17] because the gaps between the tiles usually vary in width. Therefore, the boundary conditions of a tiled liner would not have been known precisely [14].

However, the flow of coolant air through these gaps was to be mimicked as it influences the combustion process. To this end, a row of holes is located at the upstream outer edge of the chamber (Fig. 2). Here, an additional air flow termed leakage air (L) can be injected into the combustion chamber. Figure 3 shows a comparison of the liner cooling configuration implemented in the test section and the tiled design commonly used in gas turbines.

Hence, in the current setup the combustion process and its stability margins are influenced by the heat flux through the liner and the leakage air flow. While the wall heat flux is given by material properties, design features and the coolant mass flow, the leakage air flow can be set independently.

2.1 Operating Conditions. The thermodynamic conditions of the combustion air were kept constant throughout the campaign: The burner air volume flow and the burner air inlet temperature (673 K) were set to match gas turbine conditions. As seen from previous tests, the leakage air flow takes part in the chemical reaction [14]. Hence, a global equivalence ratio ϕ_{glob} is defined. It relates the overall fuel mass flow $\dot{m}_{B,f}$ to the sum of the burner air flow $\dot{m}_{a,B}$ and the leakage air flow $\dot{m}_L$

$$\phi_{\mathrm{glob}} = AFR_{\mathrm{st}} \cdot \frac{\dot{m}_{B,f}}{\dot{m}_{B,a} + \dot{m}_L} \tag{1}$$

Specific operating conditions were set using the relative pilot fuel flow rate θ and the relative leakage air flow rate L as additional variables

$$\theta = \frac{\dot{m}_{P,f}}{\dot{m}_{B,f}}, \quad L = \frac{\dot{m}_L}{\dot{m}_{B,a}} \tag{2}$$

Please note that θ relates the pilot fuel mass flow $\dot{m}_{P,f}$ to the entire fuel flow rate $\dot{m}_{B,f}$ and the leakage air flow rate $\dot{m}_{L,a}$ is nondimensionalized using the burner mass flow rate $\dot{m}_{B,a}$. These definitions allow to set the variables ϕ_{glob}, θ, and L independently.

3 Experimental Methods

Using the setup, as described in the last section, measurements of the velocity and turbulence levels for the axial and tangential velocity components (u,v) were carried out in a central planar cut through the combustor $(y=0)$. Along with this planar information, the wall heat flux, the emissions, and the oscillating and stationary pressure components were monitored.

3.1 Velocity Measurements by LDV. A standard two-dimensional (2D)-LDV setup with back-scattering receiving op-

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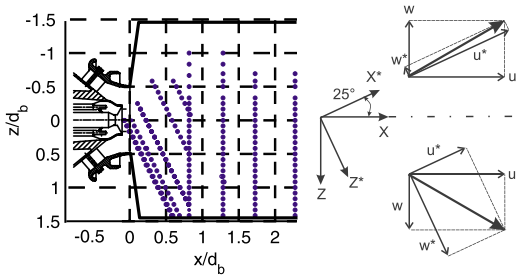

Fig. 4 Measurement locations and coordinate system for LDV.

Table 1 Operating conditions for LDV and scalar results

Operating conditions				Results		
N	ϕ_{glob}	θ (%)	L (%)	$\frac{Q_W}{Q_{ref}}$ (%)	$\frac{p_{rms}}{p_{rms,A2}}$ (%)	$\frac{\Delta p}{p_{ref}}$ (%)
A1	0.525	0	0	7[a]	1.45[a]	3.3[a]
A2	0.525	6	0	9	1	3.5
A3	0.516	0	5	6	1.42	3.3
A4	0.516	6	5	9	1.05	3.5

[a]For $A1_N$.

tics was used. Albrecht et al. [18] give a comprehensive overview on the experimental techniques involved in laser Doppler velocimetry (LDV). Details on appropriate seeding can be found in Ref. [19]. The focal length of the system used in this study was 800 mm. The size of the measurement volume was 2.1 mm in the direction of the optical axis and 120 μm in directions perpendicular to the optical axis. Seeding of the whole air flow was achieved with magnesium oxide particles, which were fed into the plenum upstream of the burner using a brush-type particle generator. Due to design features of the burner, the flow past the axial swirler received considerably less particles compared with the main airflow passing through the diagonal swirler. Using this system, data rates of up to 5 kHz were achieved for coincided measurements of the two velocity components in the main flow zone at reasonable signal-to-noise levels. Optical access was obtained by placement of a water-cooled window system in the combustor wall located at $z/d_b=1.46$. The window could be mounted along a slit, which was introduced into the whole length of the metallic liner structure giving access to one line of radial positions at a fixed axial position. This setup allowed the optical access to be comparatively small in size. The wall surface area covered by the optical access was quadratic and had an edge length of $1/4\cdot d_b$. The unused parts of the slit were sealed using quasi-adiabatic ceramic bricks.

Figure 4 shows a schematic of the burner, the combustion chamber, and the locations where velocity measurements were taken. For the measurements in the region close to the burner mouth, the optical axis was tilted by 25 deg with respect to the radial direction. As a result, instead of measuring the axial velocity component $u(t)$, the velocity $u^*(t)$ is measured in this region. It differs slightly from the axial velocity. As only two components of the velocity vector are measured, it is impossible to compute $u(t)$ from the measurement of $u^*(t)$ and $v(t)$. Hence, the results derived from axial velocity measurements close to the burner mouth ($x/d_b<0.8$) are based on the velocity component $u^*(t)$, as sketched in Fig. 4. Due to the distortion, the measured mean velocity $u^*(x,0,z)$ is smaller than the true axial velocity component $u(x,0,z)$ for $z>0$. LDV measurements for $z<0$ were carried out to prove that the flow field is axis-symmetric with respect to the burner axis. This was shown for the mean velocity components. Statistical moments of higher order were not calculated for $z<0$ as the data rate in this region was relatively low due to distortion and absorption of the laser light in the flame.

3.2 Emissions. For emission testing, flue gas was sampled by a probe. The probe inlet was situated at the center of the exit plane of the combustion chamber. A nondispersive infrared (NDIR) detector was used for the dry measurement of the concentration of carbon monoxide. A flame ionization detector was used to measure the concentration of unburnt hydrocarbons in the wet gases, while nitrogen oxide emissions were detected by chemiluminescence of NO_2.

3.3 Wall Heat Transfer. As the heat flux through the ceramic parts of the liner is considerably smaller than the flux through the metallic liner, only the latter was measured.

The surface-integrated wall heat flux Q_W was deduced from the enthalpy rise of the coolant air flow. To this end, the coolant duct was insulated with respect to the ambient. Thus, the conductive heat flux past the metallic liner equals the enthalpy rise of the coolant air. This enthalpy rise was calculated from temperature measurements using thermocouples, which were installed in the coolant duct up- and downstream of the impingement cooling system. To limit influences of inhomogeneities in the coolant flow field, the temperature and velocity fields in the coolant duct were simulated numerically and the thermocouple positions were chosen accordingly.

Values of the heat flux presented here are normalized using a constant heat flux Q_{ref}, which represents the heat released due to the chemical reaction under nominal conditions (A2 in Table 1).

3.4 Chemiluminescence. The measurements of the chemiluminescence of the CH* radical reported here were obtained using an enlarged optical access and represent the time average of line of sight images. Light was filtered with a 432 nm bandpass filter. A detailed description of the optical access used and of similar measurements was published recently [14].

4 Flow Structures

The measurements to be discussed in this section were carried out at four sets of operating conditions as listed in Table 1. (Measurement results shown in Table 1 will be discussed in Secs. 5 and 6.) The parameters were chosen to represent a variation in the pilot fuel flow (θ) and the leakage air flow (L) close to the lean blow off limit. The numbering of the sets of operating conditions reflects increasing complexity of the operating conditions: Case A1 is a nonpiloted premixed flame without leakage air input. Its piloted counterpart (A2) also has no leakage air input. It is considered the nominal operating point as it is closest to the conditions found in the quasi-adiabatic test rig studied previously [11]. The operating points with leakage air input are termed A3 (not piloted) and A4 (piloted).

The flow fields for conditions with leakage air (A3, A4) were found to show less particularities. Therefore, they will be discussed first. Differences between the nonpiloted (A3) and the piloted case (A4) will be highlighted in detail. Next, flow fields for conditions without leakage air (i.e., A1 and A2) will be presented. To ease the comparison of the flow field data, the contour plots shown here will contain data for two sets of operating conditions each. While data sets for nonpiloted cases were plotted directly on the positive z-axis, data sets for the respective piloted cases were mirrored to be shown on the negative z-axis of the same plot. Again, all data sets to be shown will depict values measured for positive z-values.

The axial mean velocity component u/u_{ref} is shown in Fig. 5(a) for A3 and A4. Normalization was carried out using the bulk axial velocity at the burner mouth (u_{ref}). In case of the piloted flame (A4), the flow emerging from the burner mouth diverges toward the combustor liner forming a relatively wide cone in which the main flow is located. In consequence, a strong and large inner recirculation zone (IRZ) is established and a relatively small outer

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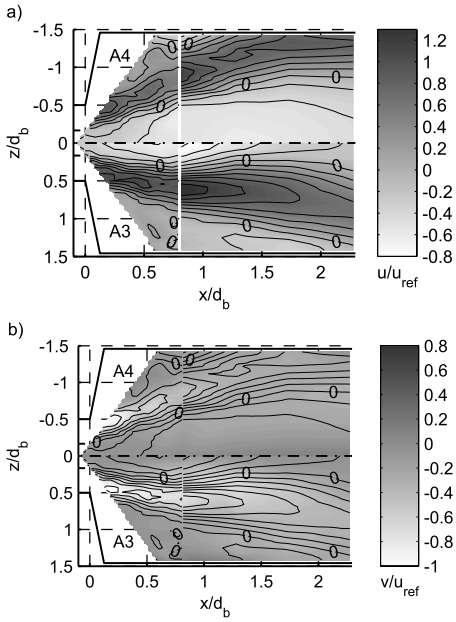

Fig. 5 Normalized mean velocity for A3 and A4 (ϕ_{glob}=0.516, L=5%): (*a*) axial and (*b*) tangential component. Isolines represent *u* in both plots

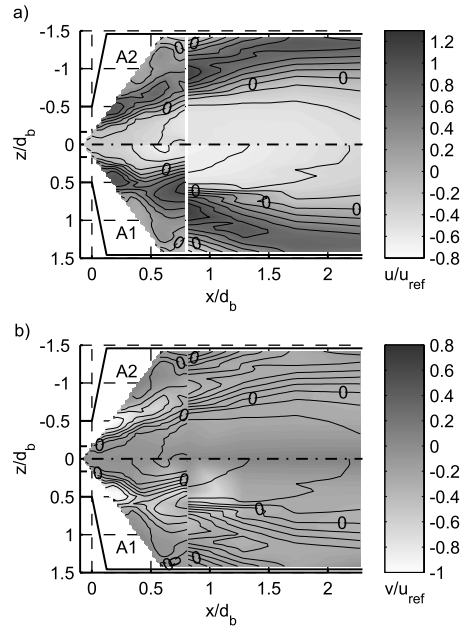

Fig. 7 Normalized mean velocity for A1 and A2 (ϕ_{glob}=0.525, L=0%): (*a*) axial and (*b*) tangential component. Isolines represent *u* in both plots.

recirculation zone (ORZ) can be inferred. Please note that the main flow zone becomes attached to the metallic liner. This flow structure was also present in the quasi-adiabatic equivalent of this test rig as reported previously [8]. In contrast, the nonpiloted case (A3) shows a significantly different flow field: The swirling main flow does not clearly break down. It forms a very narrow cone angle and is not attached to the combustor liner. Consequently, the ORZ is relatively large. Still, the ORZ does not exhibit high velocities. The IRZ is limited in size, and the absolute velocities in the IRZ are small with respect to the piloted case. This flow structure was not found as a stable operating condition in the quasi-adiabatic test rig [8].

The corresponding plots of the tangential mean velocity are given in Fig. 5(*b*). The sense of rotation is counterclockwise in all cases. As the flow does not significantly diverge in the nonpiloted case, the absolute tangential velocities remain comparatively high in the main flow zone. It can be seen that the flow field of the nonpiloted flame features a relatively wide main flow zone with velocity gradients being considerably smaller than in the piloted case.

To illustrate turbulence and possible oscillations in the flow field, the rms-value of the axial velocity is shown in Fig. 6. The plot contains lines of constant mean axial velocity for orientation. Because of the low seeding densities in the flow downstream of the axial swirler, the rms-values close to the burner axis are hard

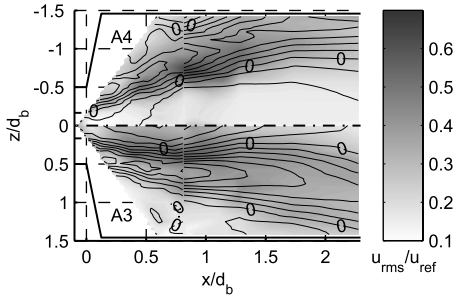

Fig. 6 rms of axial velocity u_{rms} for A3 and A4 (ϕ_{glob}=0.516, L=5%) with isolines of *u*

to interpret. Nonetheless, it can be seen from Fig. 6 that turbulence levels are highest in the shear zone between the main flow and the IRZ in both the piloted and nonpiloted case. The nonpiloted case (A3), however, exhibits an area of increased turbulence levels which matches the region of increased tangential velocity, as shown in Fig. 5(*b*).

In summary, the cases in which significant leakage air is fed into the combustion chamber exhibit two distinctively different flow fields. To ease further discussion, the two types of flow fields will be termed "wide" (A4) and "narrow" (A3) according to the shape of the main flow zones.

The measurements reported were repeated for increased leakage air ratios of L=10%. Again, wide and narrow flow fields were detected depending on the pilot fuel ratio. The flow fields were similar compared with their counterparts with L=5% and will not be discussed in further detail.

The last set of LDV measurements in the entire center plane shown here depicts the flow fields without leakage air (L=0, Fig. 7). For the piloted case (A2), a wide flow field was detected. Its main flow features are similar to the respective case with leakage air (A4). Differences between (A2) and (A4) are restricted to small zones only and are attributed to local aerodynamic and thermal effects of the leakage air input. For A1, nonuniform results were found, as displayed in the lower half of Fig. 7. Velocity measurements for $x/d_b > 0.8$ were taken early in the course of the measurement campaign. According to this subset of the data, a wide flow field is present for the pure premixed flame (termed $A1_W$). In contrast, later measurements carried out for $x/d_b < 0.8$, indicate that the flow stabilizes in the narrow shape, as known from A3 (Fig. 5(*a*), termed $A1_N$). Changes in natural gas composition or absolute humidity of the combustion air might be the reason for this change in flow pattern but were not monitored during the campaign. This high susceptibility of the flow pattern on minor changes in system parameters will be addressed in Sec. 5.

5 Transition of Flow Patterns and LBO

As two fundamentally different types of swirl flow patterns were found for the four operating conditions, the transition of the flow field between these patterns was studied. To this end, the

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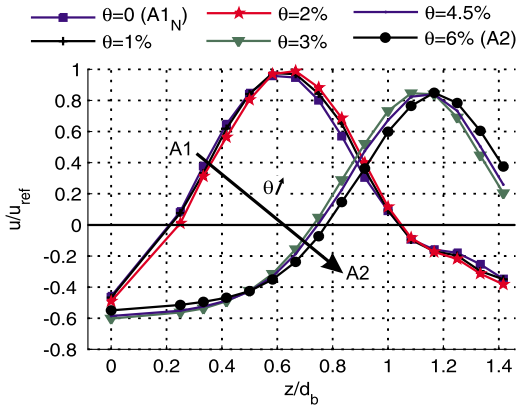

Fig. 8 Change of flow field with θ: u/u_{ref} at x/d_b=1.3 for a transition from A1$_N$ to A2

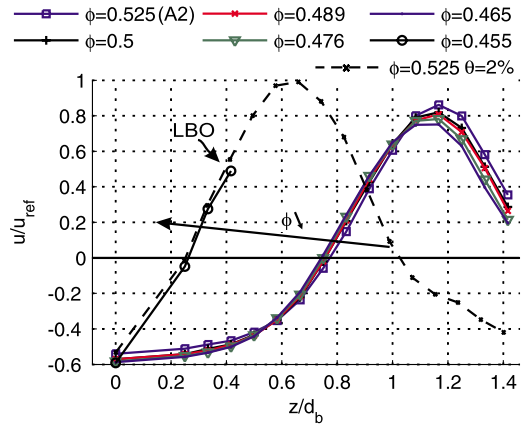

Fig. 10 Change of piloted flow field with ϕ: u/u_{ref} at x/d_b=1.3 for reduced fuel mass flow (θ=6%, L=0. Values for (ϕ=0.525, θ=2%, L=0%) plotted for reference.

pilot gas ratio θ was gradually changed in steps of 1% to obtain a transition from A1$_N$ to A2 and from A3 to A4 and vice versa.

Each step change in any operating parameter resulted in changes in heat flux and surface temperatures. These transient processes were allowed to settle by waiting for a minimum of 10 min after each step change. Monitoring of the flow field was carried out by measurement of the radial velocity profile at x/d_b=1.3.

For increasing pilot fuel flow rates, and no leakage air, the transition from A1$_N$ to A2 is shown in Fig. 8. Profiles of the mean axial velocity u/u_{ref} are given for the positive z-axis (radial coordinate). For A1$_N$, the peak mean axial velocity is situated at radial positions close to the burner outer radius (z/d_b=0.6), which clearly indicates the narrow flow pattern (A1$_N$). An increase in pilot fuel ratio θ does not significantly change the flow pattern up to θ=2%. Subsequently, diverting only 1% of the overall fuel input from the premixed flame to the pilot fuel makes the flow pattern shift dramatically from the narrow to the wide shape. Maximum axial velocities are now located at $z/d_b \approx 1.1$. A further increase in ϕ up to ϕ=6% results in only slight changes of the flow field.

The transition between the two flow patterns occurred suddenly during the course of the experiment and was tracked by the rig's monitoring system, as shown in Fig. 9. While the wall heat flux Q_W reacts with a delay caused by thermal inertia, the combustor pressure drop Δp exhibits a step change. At this moment, a change in noise emissions of the flame was also audible. Please refer to Table 1 for typical steady-state values for Δp and Q_W.

The behavior discussed for the transition from A1$_N$ to A2 was found to be typical to all those measurement series, which showed a change in flow pattern *and* did not show lean blow off right after the change. No operating condition was found to exhibit successive changes from one flow pattern to the other. Still, at different days or under different wall boundary conditions (i.e., transient boundary conditions), certain operating points exhibited one or the other flow pattern, as discussed for A1 in Sec. 4. It was found

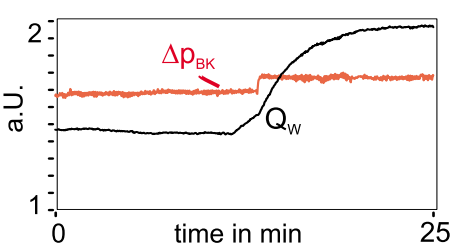

Fig. 9 From the narrow to the wide flow pattern: time trace of combustor pressure drop and wall heat flux

that all transitions were reversible by setting the boundary conditions to the values present at the start of the experiment. Due to the significant thermal inertia of the system, an investigation of possible hysteresis effects was not carried out.

The flow pattern was also found to depend on the equivalence ratio: In the nonpiloted cases (A1$_N$ and A3), the narrow flow pattern was transformed into the wide flow pattern by increasing the fuel input. Under piloted conditions (A2 and A4), the wide flow pattern was seen to change into the narrow shape for reduced overall fuel flow.

However, this transition did not always result in stable combustion. Events of LBO took place immediately after or within several minutes after the transition. An example is shown in Fig. 10. At the beginning of the experiment, the flame was operated with pilot gas at condition A2 (wide flow pattern). In the course of the experiment, the total fuel input was reduced in steps. After each step pointwise LDV measurements were carried out starting at the burner axis (z/d_b=0). This process was repeated without significant changes in the flow field until ϕ=0.465 was reached. This represents a reduction of 12% in the fuel flow rate. Then a further reduction of only 2.2% of the fuel flow triggered the transition to the narrow flow field. However, this state of the flame was not stable and LBO occurred $\approx$3 min after the transition. In this period of time, the measurement of the velocity profile could not be completed for all locations. So, velocity values are available for $z/d_b < 0.5$ only. These values fit well to the profiles of the narrow flow field, as plotted Fig. 9. For comparison, a velocity profile for a richer mixture with reduced pilot fuel, as presented in Fig. 9, is reproduced in Fig. 10.

Hence, in the case discussed in Fig. 10, LBO was preceded by a change from the wide flow field to the narrow field. In general, LBO was only observed in combination with the narrow flow field being present immediately before LBO. If the fuel supply was reduced at operating points, which exhibit the wide flow field, a transition to the narrow flow field preceded LBO. Evidently, this behavior cannot be tracked if the gradients of the transient processes are too large.

It was concluded that the narrow flow field represents a less stable flame. This is supported by measurements of the sound pressure in the plenum upstream of the burner $\tilde{p}$ (The measurement location is shown in Fig. 2). The rms-value of the sound pressure p_{rms} for the narrow flow field was found to be $\approx$40% higher than typical values for the wide flow field. This increase in sound emissions was due to weak thermo acoustic oscillations in a frequency band of 100–200 Hz superimposed to broadband noise in a frequency band of 20–500 Hz. Hence, the narrow flow field was found to be affected by acoustics and stochastic effects. In contrast, the introduction of leakage air had a significantly smaller

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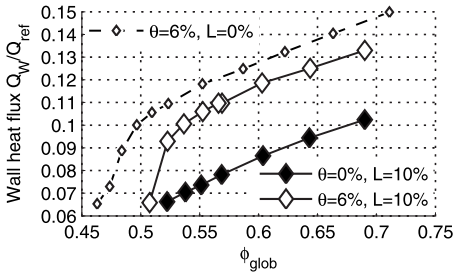

Fig. 11 Wall heat flux as a function of ϕ_{glob}

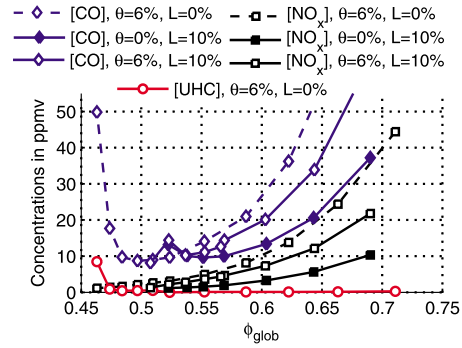

Fig. 12 Emissions of CO, NO_x, and UHC as a function of ϕ_{glob}

effect on noise. Noise levels of piloted flames operated at $L=10\%$ are only 16% higher than the equivalent flame without leakage air. A comparison of typical sound pressure levels is included in Table 1.

6 Wall Heat Flux and Emissions

Along with the LDV measurements, emissions of CO, NO_x, and unburnt hydrocarbons (UHC) as well as the wall heat flux Q_W were recorded. It was found that the wall heat flux is a reliable indicator for the flow pattern in the combustor. Generally, wall heat flux is influenced by the adiabatic flame temperature (i.e., ϕ) and the flow field, which determines the temperature distribution in the combustor: Fig. 11 shows the wall heat flux for three series of lean blow off tests at constant pilot and leakage ratios. The equivalence ratio was set to relatively rich levels and was reduced in steps until the flame was extinguished. For large equivalence ratios, the heat flux was approximately proportional to the equivalence ratio for all cases. This is attributed to the influence of the equivalence ratio on the temperature level in the combustion chamber. Furthermore, the piloted cases shown in Fig. 11 exhibit a higher heat flux compared with the nonpiloted case, which is explained by the main flow being attached to the liner in a wide flow field. Comparison of the heat flux values for the two piloted test series presented in Fig. 11 reveals that injection of cold leakage air into the ORZ reduces the wall heat flux. This is attributed to the gas temperatures in the vicinity of the liner being reduced due to the injection of leakage air.

For equivalence ratios close to LBO the heat flux decreases dramatically when reducing ϕ for the piloted cases. It drops to values similar to the heat flux present for the nonpiloted case close to LBO. The behavior of the piloted flames is attributed to a transition of the flow field from wide to narrow. Hence, the wall heat flux is considered a good indicator for imminent lean blow out. This could be explored in practical applications as the instrumentation needed does not have contact with the hot gases. Furthermore, the case without leakage air exhibits a leaner LBO limit compared with the cases with $L=10\%$. However, this behavior could not be reproduced to a satisfactory extent.

Coincident with the wall heat flux measurements presented in Fig. 11 are the emissions tests, as shown in Fig. 12. UHC concentrations were below the detection limit, except for the piloted case without leakage air, where the time traces of UHC concentrations showed spikes. The resulting increase in the mean value for lean mixtures is given in Fig. 12. For the same case, CO-concentrations exhibited high emissions for extremely lean conditions, as reported in the field tests [7]. In cases with significant leakage air, CO-concentrations were found to show only a slight increase close to the lean blow off limit. Close to LBO, CO-emissions did not necessarily show an increase as it might be expected to be triggered by local quenching due to low flame temperatures. Increases of nitrogen oxide concentrations were detected for increasing flame temperatures in all cases, which is in accordance with the Zeldovich-mechanism of NO_x-formation. As shown in Fig. 12 the measured emissions of CO and NO_x were not independent of piloting and leakage air. As expected from field tests [7], the NO_x generation was found to be promoted by pilot fuel injection.

7 Flame Structure

Changes in the position and shape of the flame were also observed using the chemiluminescence of OH and CH radicals. The observations discussed here were evident from both the OH and CH-chemiluminescence signals. Only the latter set of information shall be discussed.

Figure 13 shows the time-averaged CH-chemiluminescence of the flame close to the burner mouth. The burner and the ceramic liner of the combustor back wall are visible in the images because the hot combustor walls also emit light at the wavelength of the chemiluminescence signal. It can be seen that the nonpiloted flame emits a comparatively weak chemiluminescence signal near the burner. In particular, the piloted flames showed a conical structure based at the axial swirler outlet, which was not visible for any nonpiloted flame. This conical structure might be called a "pilot flame." Still, there is no evident boundary separating this structure from the main combustion zones. For all cases, the main combustion zones were detected downstream of the field of view depicted in Fig. 13. They showed higher intensities for piloted cases. It is concluded that piloting shifts parts of the heat release to the conical zone close to the burner. At the same time, the volume of the zone with high heat release (i.e., flame) is reduced. Quantitative investigations of the heat release as a function of piloting and leakage air based on measurements of the temperature field in the combustion chamber will be presented in the near future. These data sets will be interpreted together with the data presented to allow for a more detailed explanation of the effects described here.

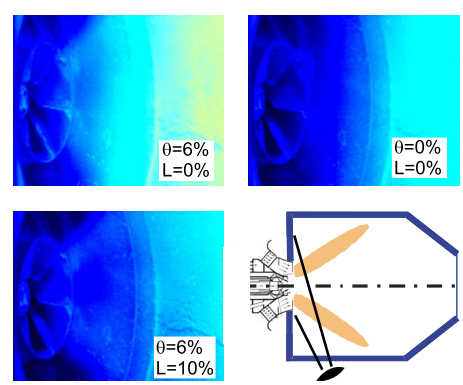

Fig. 13 CH*-chemiluminescence at the burner outlet for ϕ=0.5 and field of view

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8 Conclusion

It was demonstrated that the positive effects of pilot fuel injection on the stability and the lean blow off limit of an industrial lean premixed combustion system can be reproduced using a distinctively nonadiabatic test setup. A similar but quasi-adiabatic combustion system tested earlier [13] did not show effects of piloting on the flame properties. As the current nonadiabatic test rig exhibits such effects, the importance of wall heat flux for investigations on stability of lean premixed flames is emphasized.

The shape of the reacting flow field within the nonadiabatic combustor was severely affected by pilot fuel injection. Two flow patterns were detected. They differ in the width of the cone angle of their main flow zone and the resulting size of the inner recirculation zone. The wide flow pattern occurred at high load, sufficient piloting, and/or reduced leakage air flow. It did coincide with a more stable and more silent combustion process. For lower load, reduced piloting and/or increased leakage air, a sudden transition to the narrow flow field was observed. This transition results in changes of the pressure loss in the system, the wall heat flux and the pollutant emissions. The studies revealed that the wall heat flux is an easily accessible indicator for the shape of the flow pattern and hence for the stability of the flame. It was seen from chemiluminescence imaging that piloting shifts the reaction zones toward the burner. As lean blow off did only occur if the narrow flow field was present before the LBO, it was concluded that the flame stability is severely affected by the change in the flow structure.

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Nomenclature

Latin Letters

AFR_{st} = air-fuel ratio for stoichiometric conditions
d_b = diameter of burner mouth (m)
L = leakage air flow ratio (Eq. (2))
$\dot{m}$ = mass flow (kg/s)
p = pressure (Pa)
Q_W = wall heat flux (W)
x,y,z = spatial coordinates (m)
u,v,w = (time average of) velocity components (m/s)
$[\ldots]$ = volumetric concentration

Greek Letters

ϕ_{glob} = global equivalence ratio (Eq. (1))
θ = pilot fuel ratio (Eq. (2))

Subscripts

a = air
b = burner
f = fuel (i.e., natural gas)
L = leakage air system
$N(W)$ = cone of main flow narrow (wide)
P = pilot
ref = reference value
rms = root mean square

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